

TUCSON, AZ, USA

WMO: 722740

Lat: 32.228N

Lon: 110.956W

Elev: 747

StdP: 92.66

Time Zone: -7.00 (W07)

Period: 94-19

WBAN: 23160

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	0.0	1.5	-18.4	0.8	16.8	-16.1	1.0	17.3	11.0	15.2	9.7	15.0	2.2	140	0.462

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	40.9	18.9	39.8	18.7	38.6	18.6	22.6	31.0	22.1	30.7	21.8	30.5	4.6	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	16.9	24.8	20.1	16.2	25.0	19.4	15.5	25.5	70.0	30.6	68.5	30.5	66.9	30.1	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.5	8.3	7.4	-3.1	43.2	2.0	1.4	-4.5	44.3	-5.7	45.1	-6.8	45.9	-8.2	46.9
			-5.7	23.5	2.0	0.6	-7.2	24.0	-8.3	24.4	-9.4	24.7	-10.9	25.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.4	12.0	13.4	16.8	20.0	24.7	30.2	31.1	30.2	28.1	22.3	16.5	11.6
	DBStd	7.80	3.49	3.67	3.76	3.85	3.36	2.73	2.41	2.20	2.35	3.70	3.99	3.57
	HDD10.0	64	21	11	3	1	0	0	0	0	0	0	4	26
	HDD18.3	738	196	142	72	28	2	0	0	0	0	11	78	209
	CDD10.0	4242	83	106	214	299	456	605	653	627	542	382	199	75
	CDD18.3	1874	1	3	25	77	199	355	394	368	292	135	24	1
	CDH23.3	26251	30	99	478	1151	2884	5574	5621	4904	3607	1535	344	23
CDH26.7	15028	2	14	130	476	1601	3663	3469	2874	2003	707	88	1	
Wind	WSAvg	3.3	3.0	3.1	3.4	3.6	3.5	3.5	3.4	3.2	3.3	3.2	3.1	2.9
Precipitation	PrecAvg	298	25	23	21	9	5	6	48	57	32	24	18	29
	PrecMax	501	142	97	60	40	21	50	192	130	112	147	45	185
	PrecMin	176	0	0	0	0	0	0	1	8	0	0	0	0
	PrecStd	86	26	23	17	10	6	11	35	31	28	32	13	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.1	28.6	32.1	34.9	39.2	43.0	42.4	41.4	39.1	36.0	31.2	25.8
		MCWB	11.1	11.5	13.3	14.2	17.0	18.6	19.1	20.1	19.0	17.1	13.8	11.2
	2%	DB	24.0	26.2	29.7	32.9	37.2	41.1	40.8	39.7	37.6	34.2	29.1	23.6
		MCWB	10.3	10.9	12.4	13.6	15.9	17.9	19.5	20.0	18.7	16.4	13.0	10.7
	5%	DB	22.0	24.3	28.0	31.4	35.6	39.9	39.5	38.3	36.5	32.7	27.3	21.4
		MCWB	9.5	10.3	11.8	13.0	15.2	17.6	19.7	20.1	18.5	15.7	12.6	9.9
	10%	DB	19.9	21.9	26.2	29.6	34.1	38.7	38.0	36.8	35.1	31.0	25.2	19.3
		MCWB	8.9	9.6	11.1	12.4	14.5	17.1	19.7	20.2	18.6	15.1	12.0	9.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.9	14.0	15.0	15.8	18.8	21.7	23.0	23.2	22.8	20.2	15.6	13.6
		MCDB	18.9	20.7	26.8	31.5	34.4	33.1	30.9	31.4	31.0	27.0	22.6	19.0
	2%	WB	12.1	12.8	13.7	14.6	17.3	20.7	22.5	22.7	21.9	18.8	14.6	12.2
		MCDB	19.3	20.5	26.1	30.0	32.9	33.4	30.6	31.2	29.2	27.7	23.9	19.8
	5%	WB	10.9	11.7	12.7	13.7	16.2	19.9	22.0	22.2	21.3	17.7	13.6	11.1
		MCDB	18.6	20.0	24.9	28.8	32.2	33.2	30.4	31.1	29.0	28.0	24.1	18.8
	10%	WB	10.0	10.7	11.7	12.8	15.3	19.0	21.7	21.8	20.7	16.7	12.7	10.1
		MCDB	18.0	19.8	24.0	27.9	31.9	34.0	30.4	30.9	28.9	27.6	23.5	17.8
Mean Daily Temperature Range	5% DB	MDBR	14.2	14.3	15.3	15.9	16.2	15.9	12.7	12.4	13.1	14.7	14.7	13.7
		MCDBR	17.2	18.0	18.1	18.0	17.9	17.1	14.6	14.4	14.9	16.6	17.1	16.3
		MCWBR	8.2	8.2	7.6	7.4	7.0	6.0	3.9	3.3	4.1	6.3	7.4	7.8
	5% WB	MCDBR	13.5	14.3	15.9	16.5	16.4	14.0	11.8	11.9	11.5	12.8	14.2	13.6
		MCWBR	6.8	6.6	6.6	7.0	6.5	4.7	2.7	2.9	3.2	4.9	6.5	6.9
Clear-Sky Solar Irradiance	taub	0.263	0.267	0.289	0.299	0.311	0.333	0.392	0.379	0.338	0.300	0.273	0.262	
	taud	2.632	2.625	2.533	2.488	2.492	2.437	2.403	2.428	2.515	2.566	2.599	2.627	
	Ebn at Noon	980	1005	997	990	974	946	889	899	930	950	958	963	
	Edh at Noon	75	84	99	108	109	115	118	114	100	88	77	72	
All-Sky Solar Radiation	RadAvg	3.59	4.53	6.02	7.40	8.19	8.22	6.86	6.48	5.93	5.05	3.97	3.28	
	RadStd	0.25	0.36	0.29	0.23	0.30	0.43	0.41	0.23	0.28	0.32	0.19	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
	+0.37	N/A	N/A	N/A	N/A	N/A	N/A	-81	+159	+68	
Regional (13 neighbors)		+0.27	N/A	N/A	N/A	N/A	N/A	-75	+104	N/A	

Nomenclature: See separate page